

Project Aura

A wearable spatial-awareness system that compresses the 3D world into touch-first guidance: a 2D projection, a depth layer, and haptic language the body can read quickly.

Seeing the environment is not the same as understanding where it is safe to move.

Vest

Ultrasonic depth and haptic reflexes

Phone

Camera, object detection, fusion, TTS

Watch

Intentional scene trigger

User

Guidance without visual attention

Hero visual placeholder

Use a vest photo, phone demo screenshot, or user journey illustration

PROBLEM

Spatial awareness has to be fast, calm, and non-visual.

PRIMARY USER

Visually impaired people need immediate obstacle awareness and optional scene understanding without staring at a screen.

DESIGN PRESSURE

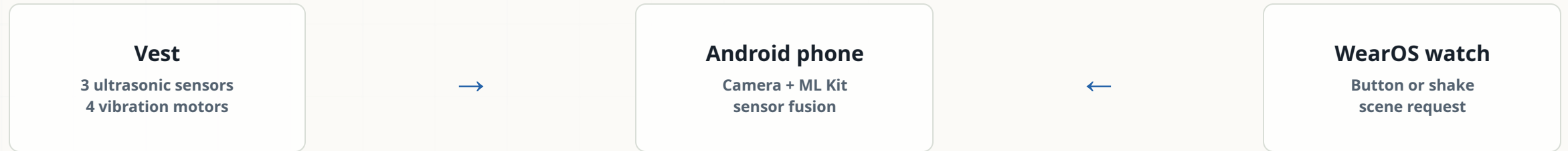
- Obstacle avoidance needs very low latency.
- Audio descriptions are useful, but constant speech is distracting.
- Visual UI cannot be the main feedback channel.

Problem context visual

Add photo of navigation scenario, obstacle field, or interview insight

SOLUTION

Aura splits spatial perception into reflexes, intelligence, and intent.



Reflex layer

The vest runs autonomously. If the phone is disconnected, distance-to-vibration still works.

Interpretation layer

The phone recognizes objects, maps them to zones, fuses them with depth, and sends haptic overrides.

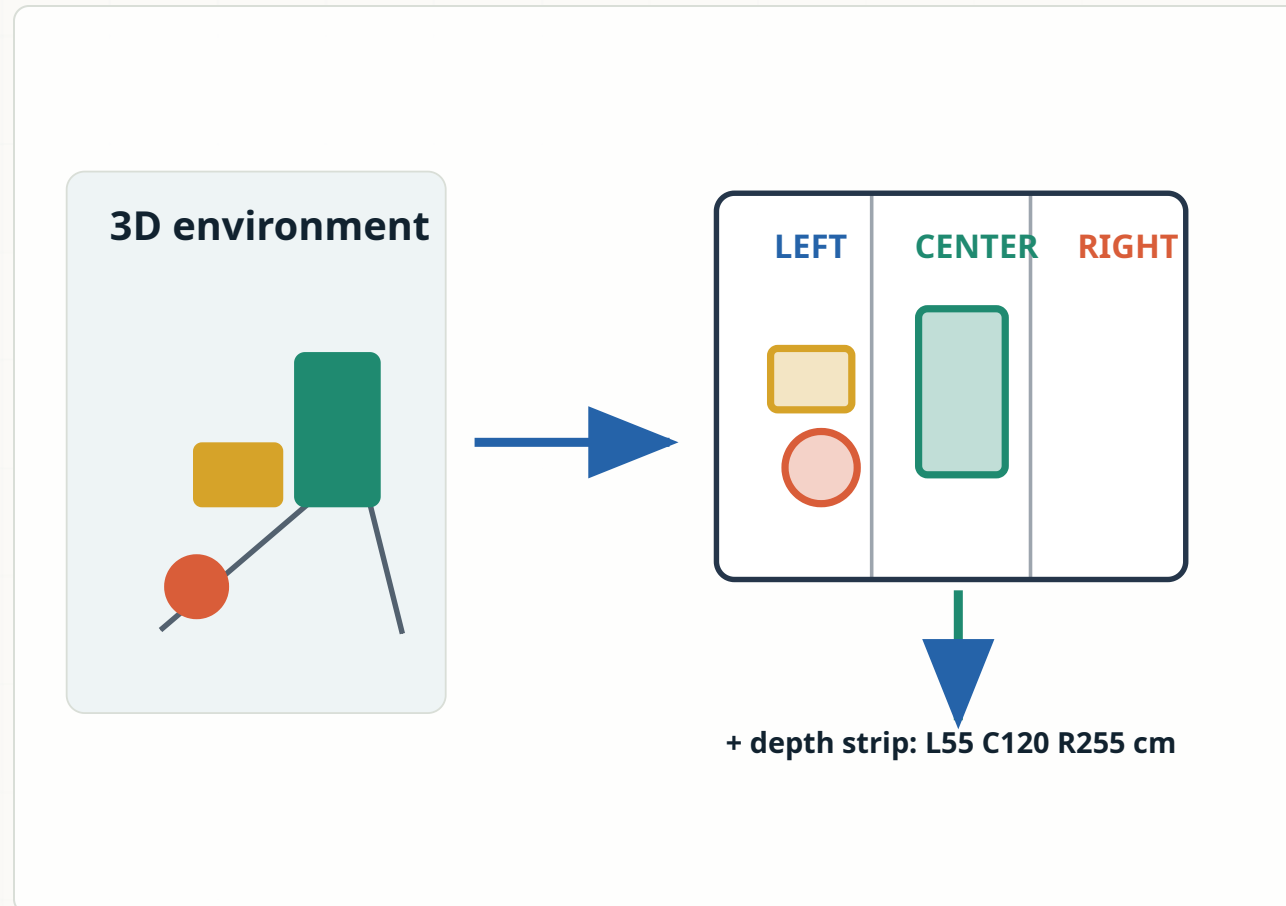
Intent layer

The watch asks for a spoken scene description only when the user wants more context.

Current transport: vest over WiFi HTTP, watch over BLE

CORE IDEA

We do not need a full 3D reconstruction. We need an actionable projection.



3D world -> 2D zones + depth

Compression, not simulation

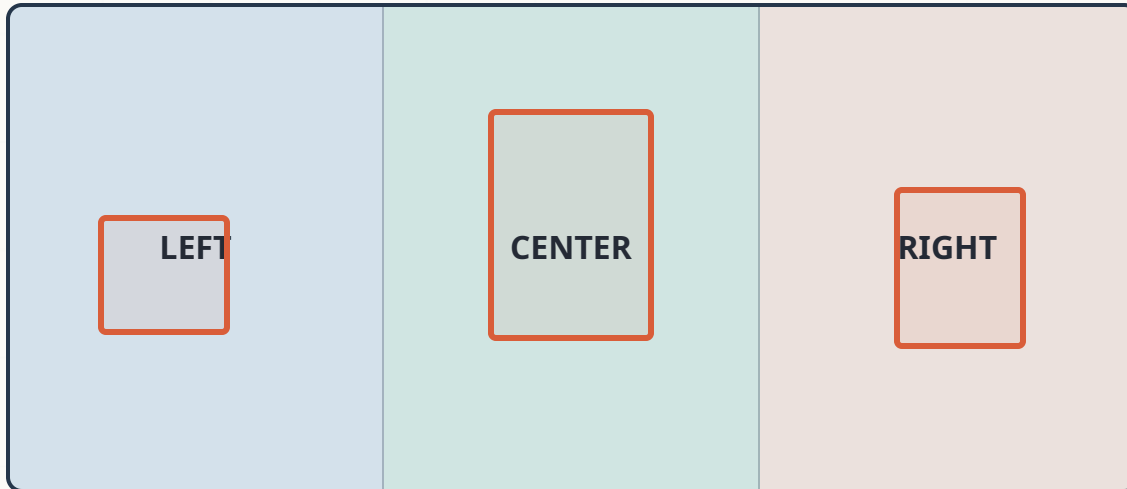
The phone camera gives an image plane. ML Kit places detected objects in that plane. The vest supplies real distance per body zone. Aura combines them into: **what, where left-center-right, and how close.**

Presentation line: "We turn a messy 3D scene into a compact 2.5D signal the user can feel."

Projection illustration placeholder

Add photo-to-grid annotation or hand-drawn sketch

The 2D image is split into body-aligned navigation zones.



```
zone = bbox.centerX < 0.33w ? left : bbox.centerX > 0.66w ?  
right : center
```

This makes object positions match the body: left vibration for left-side risk, center vibration for forward path, right vibration for right-side risk.

2D screen space becomes tactile body space

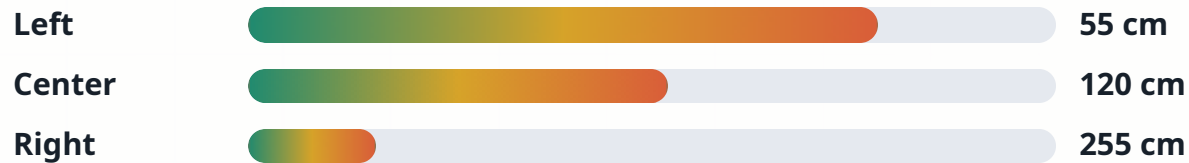
Camera detection screenshot placeholder

Use a phone screenshot with bounding boxes and left/center/right overlay

Ultrasonic sensors add the missing dimension: distance.

VEST SENSOR PAYLOAD

Three distances are treated as a depth strip across the user's body: left, center, right.



Why this matters: camera boxes say "object in center"; depth says whether it is safe background or near-path danger.

Depth visualization placeholder

Add diagram of 3 HC-SR04 cones or live sensor values from the app

The fusion step decides the strongest useful signal per zone.

1

Detect objects in the camera image.

2

Classify each object as left, center, or right.

3

Read vest depth in the matching zone.

4

Send the strongest non-confusing haptic command.

Example

A person appears in the center of the camera frame. If center depth is near, Aura sends a strong center warning. If the vest already reports immediate danger, Aura avoids duplicate conflicting feedback.

Result

The user receives one clear tactile message per zone: static, dynamic, danger, or clear. The phone can still speak object details on request.

Feedback is encoded as location, intensity, and pattern.

SIGNAL	USER MEANING	MOTOR BEHAVIOR
Zone	Where the risk is	Left, center, right, or both center motors
Intensity	How urgent it feels	PWM strength, capped for motor safety
Pattern	What kind of object or event	Static, heartbeat/dynamic, continuous danger

The body reads vibration faster than a sentence.

That is why Aura keeps obstacle avoidance haptic and leaves detailed speech as an intentional action.

Haptic demo visual placeholder

Add vest motor layout photo, waveform diagram, or tactile pattern illustration

The vest remains useful even when everything else disconnects.

DISTANCE	VIBRATION	MEANING
> 200 cm	Off	Clear path
150-200 cm	Slow pulse	Distant awareness
100-150 cm	Medium pulse	Approaching obstacle
50-100 cm	Fast pulse	Near obstacle
20-50 cm	Rapid pulse	Very close
< 20 cm	Continuous	Immediate danger

Implementation note: current firmware caps motor duty for safe driving from the battery and L293D setup.

Fail-safe reflexes first

Bench test visual placeholder

Add video frame or photo of hand approaching sensors and motors responding

The watch turns object detection into an intentional scene description.

User action

Press the watch button or shake the wrist. The watch vibrates briefly to confirm the trigger.

Phone response

The Android app speaks the latest scene: grouped by left, center, right, and enriched by proximity warnings.

"Person center. Chair left. Obstacle very close on right."

Why not always speak? Audio is powerful, but haptics preserve attention and keep the environment audible.

Always-on touch, optional voice

Watch and TTS screenshot placeholder

Add WearOS button screen plus Android last-scene panel

BUILD

Prototype architecture: real sensors, real app, real feedback loop.

Vest firmware

NuttX on Sparrow / ESP32 with HC-SR04 depth sensors.

Haptics

Four motors through L293D, controlled by HTTP override payloads.

Android app

ML Kit detection, fusion, TTS, watch BLE, and vest WiFi.

WearOS trigger

Large scene-request interaction with haptic confirmation.

Built as modules: firmware, mobile app, and watch trigger can be tested independently.

Hardware/software collage placeholder

Add board wiring, Android status UI, watch UI, and test setup

A simple scenario shows the whole loop.

Setup

Phone connects to vest URL and subscribes to watch trigger.

Approach

A hand or obstacle moves through left, center, right depth zones.

Recognize

Camera detects a person or object and assigns a zone.

Feel

Vest vibrates the matching zone; watch trigger speaks the scene.

Screenshot 1

Vest state values

Screenshot 2

Camera detection

Screenshot 3

Recommended live demo path

Aura makes spatial information ambient instead of demanding attention.

What is novel here?

The system treats 3D perception as an interface problem. Aura is a spatial translator: it turns the world in front of the user into a tactile map they can trust.

Fast

Vest haptics respond without waiting for speech or UI.

Calm

Speech happens only when requested.

Resilient

The vest still works in haptic-only mode.

Readable

Left, center, right, and depth become a body-aligned signal.

Final impact visual placeholder

Add user story photo, polished product render, or before/after diagram